Infinite Series

9.1 Sequences

Example:

① $1, 2, 3, 4, 5, \ldots$ ←

② $\frac{1}{1}, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \ldots$

Example: what is the next number?

$a_1 \quad a_2 \quad a_3 \quad a_4 \quad a_5 \quad a_6 \quad a_7 \ldots$

$1, \quad 2, \quad 3, \quad 4, \quad 5, \quad 6, \quad 7, \ldots$

(*) Next number could be 8 if the sequence's rule is: $a_n = n$

(*) $a_n = (n-1)\cdot(n-2)\cdot(n-3)\cdot(n-4)\cdot(n-5)\cdot(n-6)\cdot(n-7) + n$

$a_1 = 1, \quad a_2 = 2, \quad \ldots \quad a_7 = 7$

$a_8 = 7\cdot 6\cdot 5\cdot 4\cdot 3\cdot 2\cdot 1 + 8 = 5048$

(*) $a_n = (n-1)^2 \cdot (n-2) \cdot (n-3)^6 \cdot (n-4) \cdot (n-5) \cdot (n-6) \cdot (n-7)^7 + n$

To define a sequence we need to state the rule clearly.

(1) $a_n = 2n + 1$

$\Rightarrow \{a_n\} = 1, 3, 5, 7, \ldots, 2n+1, \ldots$

(2) $a_n = \{2n+1\}_{n=1}^{\infty}$

Definition: The sequence $\{a_n\}$ is a function of n where n is positive integers.

Regular function: input x

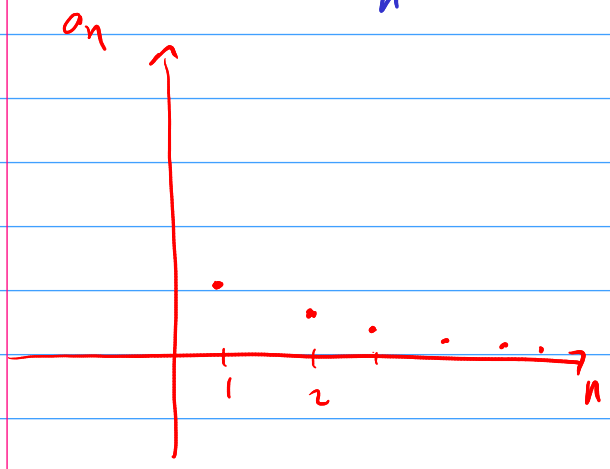
output y

Sequence: input n

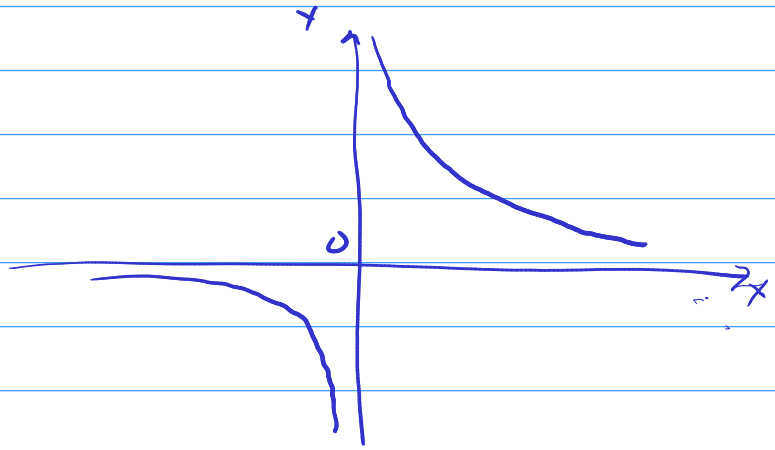
output a_n

(*) Graph of sequences

$a_n = \frac{1}{n}$



Function $y = \frac{1}{x}$



(*) Limit of sequences

Example ① $a_n = \frac{1}{n}$

$$\lim_{n \to \infty} a_n = \lim_{n \to \infty} \frac{1}{n} = 0 \quad \text{(convergent)}$$

② $a_n = 2n + 2$

$$\lim_{n \to \infty} a_n = \lim_{n \to \infty} (2n + 2) = \infty$$

This sequence is divergent

9.1.3 THEOREM *Suppose that the sequences $\{a_n\}$ and $\{b_n\}$ converge to limits L_1 and L_2, respectively, and c is a constant. Then:*

(a) $\lim_{n\to+\infty} c = c$

(b) $\lim_{n\to+\infty} ca_n = c \lim_{n\to+\infty} a_n = cL_1$

(c) $\lim_{n\to+\infty} (a_n + b_n) = \lim_{n\to+\infty} a_n + \lim_{n\to+\infty} b_n = L_1 + L_2$

(d) $\lim_{n\to+\infty} (a_n - b_n) = \lim_{n\to+\infty} a_n - \lim_{n\to+\infty} b_n = L_1 - L_2$

(e) $\lim_{n\to+\infty} (a_n b_n) = \lim_{n\to+\infty} a_n \cdot \lim_{n\to+\infty} b_n = L_1 L_2$

(f) $\lim_{n\to+\infty} \left(\frac{a_n}{b_n}\right) = \frac{\lim_{n\to+\infty} a_n}{\lim_{n\to+\infty} b_n} = \frac{L_1}{L_2}$ *(if $L_2 \neq 0$)*

Example : Find $\lim_{n\to\infty} a_n$

① $a_n = \dfrac{3n+1}{4n+2}$

« L'hospital Rule »

$$\lim_{n\to\infty} \frac{3n+1}{4n+2} = \lim_{n\to\infty} \frac{(3n+1)'}{(4n+1)'} = \frac{3}{4}$$

② Squeezing Theorem.

Suppose : $a_n \leq b_n \leq c_n$

If $\lim_{n\to\infty} a_n = \lim_{n\to\infty} c_n = K$

then $\lim_{n\to\infty} b_n = K$

Example: $a_n = \dfrac{\sin^2 n}{n}$

$$\underset{\downarrow}{\frac{0}{n}} \leq \underset{\downarrow}{\frac{\sin^2 n}{n}} \leq \underset{\downarrow}{\frac{1}{n}}$$

$$0 \qquad 0 \qquad 0$$

$$\Rightarrow \lim_{n\to\infty} \frac{\sin^2 n}{n} = 0$$

(*) $a_n = \dfrac{n!}{n^n}$

$n! = 1 \cdot 2 \cdot 3 \cdot \cdots \cdot n$

$n^n = n \cdot n \cdot n \cdots n$

$$\frac{0}{n^n} \leq \frac{n!}{n^n} = \frac{1 \cdot 2 \cdot 3 \cdots n}{n \cdot n \cdot n \cdots n}$$

$\leq$

0

$$= \frac{1}{n} \cdot \left(\frac{2}{n} \cdot \frac{3}{n} \cdots \frac{n}{n} \right)$$

$$= \underset{\downarrow}{\frac{1}{n}} \cdot 1 \cdot 1 \cdots 1 \quad \text{when } n \text{ large}$$

0

$$\Rightarrow \lim_{n\to\infty} a_n = 0$$

(*) Recursively defined sequences.

Example:

$$a_1 = 1$$

$$a_{n+1} = 1 + \frac{1}{a_n}$$

Find $\lim_{n\to\infty} a_n$ assuming the limit exists.

Suppose $\lim_{n\to\infty} a_n = L$

$$\Rightarrow \lim a_{n+1} = \lim\left(1 + \frac{1}{a_n}\right)$$

$$\Rightarrow \lim a_{n+1} = \lim 1 + \lim\left(\frac{1}{a_n}\right)$$

$$\lim a_{n+1} = 1 + \frac{1}{\lim a_n}$$

$$\Rightarrow L = 1 + \frac{1}{L}$$